

Integer-Valued Polynomials on Products Are Not Always Tensor Products

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Abstract

We resolve an open problem [2] in commutative ring theory: Given an integer domain D , we prove that the canonical map from $\text{Int}(D)^{\otimes_{D^n}}$ to $\text{Int}(D^n)$ need not be surjective or injective. The proof is generated by GPT 5.5 Pro and verified by Claude Opus 4.8 ultracode.

1 Introduction

Let D be an integral domain with quotient field $K = \text{Frac}(D)$, and let

$$\text{Int}(D) = \{ f \in K[X] : f(D) \subseteq D \}$$

be its ring of integer-valued polynomials (see Cahen–Chabert [1]). For an integer $n \geq 1$ there is a canonical D -algebra homomorphism

$$\theta_n : \text{Int}(D)^{\otimes_{D^n}} \longrightarrow \text{Int}(D^n), \quad f_1 \otimes \cdots \otimes f_n \longmapsto \prod_{i=1}^n f_i(X_i),$$

sending a pure tensor to the product of the f_i , viewed as polynomials in the separate variables $X_1, \dots, X_n$; here $\text{Int}(D^n) = \{ F \in K[X_1, \dots, X_n] : F(D^n) \subseteq D \}$. The map θ_n is an isomorphism when $\text{Int}(D)$ is a free D -module, and more generally when D is locally free with $\text{Int}(D_{\mathfrak{m}}) = \text{Int}(D)_{\mathfrak{m}}$ for every maximal ideal $\mathfrak{m}$. Problem 20 of Glaz’s list [2] asks whether θ_n is *always* injective, and whether it is *always* surjective. We answer both questions in the negative.

2 Main Proof

Theorem 2.1. *There is a one-dimensional Noetherian local integral domain D such that, for every integer $n \geq 2$, the canonical map $\theta_n : \text{Int}(D)^{\otimes_{D^n}} \rightarrow \text{Int}(D^n)$ is neither injective nor surjective. In particular θ_n is never an isomorphism for this D .*

The domain is completely explicit and the entire argument is self-contained: every assertion below is proved directly, with no appeal to external results beyond standard facts about localizations of a polynomial ring and the elementary theory of Int . After constructing D (§2.1) we prove a single “difference lemma” (§2.2) that drives both failures: non-injectivity (§2.3), non-surjectivity (§2.4), and the passage to all $n \geq 2$ (§2.5).

2.1 The domain

Let $A = \mathbb{F}_2[t]$ the polynomial ring in one variable. Let

$$S = A \setminus ((t) \cup (t+1)), \quad T = S^{-1}A.$$

Since (t) and $(t+1)$ are the only prime ideals of A that S avoids, T is a semilocal principal ideal domain with exactly two maximal ideals

$$\mathfrak{n}_0 = tT, \quad \mathfrak{n}_1 = (t+1)T.$$

Put

$$\pi = t(t+1), \quad \mathfrak{m} = \pi T = \mathfrak{n}_0 \mathfrak{n}_1 = \mathfrak{n}_0 \cap \mathfrak{n}_1, \quad D = k + \mathfrak{m} \subseteq T.$$

Thus D consists of the elements of T whose image in $T/\mathfrak{m}$ lies in the diagonal copy of $\mathbb{F}_2$, as we now make precise.

Proposition 2.2. *With the above notation:*

1. $T/\mathfrak{m} \cong \mathbb{F}_2 \times \mathbb{F}_2$ via the residues at $\mathfrak{n}_0, \mathfrak{n}_1$, and the image of $\mathbb{F}_2$ is the diagonal; consequently $D/\mathfrak{m} \cong \mathbb{F}_2$ and $\mathfrak{n}_0 \cap D = \mathfrak{n}_1 \cap D = \mathfrak{m}$.
2. $\text{Frac}(D) = \mathbb{F}_2(t)$.
3. D is a one-dimensional Noetherian local integral domain with maximal ideal $\mathfrak{m}$.

Proof. (1) Since $\mathfrak{m} = \mathfrak{n}_0 \cap \mathfrak{n}_1$ with $\mathfrak{n}_0, \mathfrak{n}_1$ comaximal, the Chinese remainder theorem gives $T/\mathfrak{m} \cong T/\mathfrak{n}_0 \times T/\mathfrak{n}_1$. Each residue field $T/\mathfrak{n}_i$ is a localization-quotient of $\mathbb{F}_2[t]$ by a degree-one prime, hence equals $\mathbb{F}_2$; so $T/\mathfrak{m} \cong \mathbb{F}_2 \times \mathbb{F}_2$. The nonzero constant $1 \in \mathbb{F}_2$ maps to $(1, 1)$, the diagonal unit, so the image of $k \cong \mathbb{F}_2$ is the diagonal $\{(0, 0), (1, 1)\}$. By definition $D = \mathbb{F}_2 + \mathfrak{m}$ is the preimage in T of this diagonal, whence $D/\mathfrak{m} \cong \mathbb{F}_2$. Finally, if $x \in \mathfrak{n}_0 \cap D$ then its residue lies in $\mathfrak{n}_0/\mathfrak{m} = \{0\} \times \mathbb{F}_2$ and simultaneously in the diagonal, forcing it to be 0; thus $x \in \mathfrak{m}$, i.e. $\mathfrak{n}_0 \cap D \subseteq \mathfrak{m}$, and the reverse inclusion is clear. Symmetrically $\mathfrak{n}_1 \cap D = \mathfrak{m}$.

(2) We have $\pi = t(t+1) \in \mathfrak{m} \subseteq D$ and $\pi t = t^2(t+1) \in \mathfrak{m} \subseteq D$, so $t = (\pi t)/\pi \in \text{Frac}(D)$. Hence $k(t) = \mathbb{F}_2(t) \subseteq \text{Frac}(D)$. Conversely $D \subseteq T \subseteq \mathbb{F}_2(t)$, so $\text{Frac}(D) \subseteq \mathbb{F}_2(t)$. Therefore $\text{Frac}(D) = \mathbb{F}_2(t) = \text{Frac}(T)$.

(3) D is a subring of the field $\mathbb{F}_2(t)$, hence a domain. We show $\mathfrak{m}$ is its unique maximal ideal. By (1), $D/\mathfrak{m} \cong \mathbb{F}_2$ is a field, so $\mathfrak{m}$ is maximal. If $x \in D \setminus \mathfrak{m}$, then $x \equiv 1 \pmod{\mathfrak{m}}$ (the only nonzero residue), say $x = 1 + \mu$ with $\mu \in \mathfrak{m}$. As $\mu \in \mathfrak{m} = \pi T \subseteq \mathfrak{n}_0 \cap \mathfrak{n}_1$, the element $1 + \mu$ is a unit of T with $(1 + \mu)^{-1} \equiv 1 \pmod{\mathfrak{m}}$ in T ; in particular $(1 + \mu)^{-1} \in \mathbb{F}_2 + \mathfrak{m} = D$. Thus every element of $D \setminus \mathfrak{m}$ is a unit of D , so D is local with maximal ideal $\mathfrak{m}$. Finally, $T = \mathbb{F}_2[t]_S$ is a finitely generated (indeed module-finite) D -module: $T = D + Dt$, since any $\frac{f(t)}{s} \in T$ can be written, after reducing f modulo π and clearing the unit s , in the form $a + bt$ with $a, b \in D$ using $\pi, \pi t \in D$. As T is a Noetherian domain that is module-finite over the domain D and shares its quotient field, the Eakin–Nagata theorem shows D is Noetherian, and $\dim D = \dim T = 1$. $\square$

Throughout, write $R = \text{Int}(D)$. Because $\mathfrak{m}$ is an ideal of T and a polynomial $f \in K[X]$ with $f(D) \subseteq \mathfrak{m}$ certainly has $f(D) \subseteq D$, such an f lies in R . We record the two running facts about valuations.

Let $v_0, v_1 : \mathbb{F}_2(t)^\times \rightarrow \mathbf{Z}$ be the discrete valuations attached to the prime ideals $\mathfrak{n}_0, \mathfrak{n}_1$ of the PID T , normalized by

$$v_0(t) = 1, \quad v_1(t+1) = 1.$$

Then $v_0(t+1) = v_1(t) = 0$ (both $t, t+1$ are units at the other prime), so

$$v_0(\pi) = v_1(\pi) = 1, \quad \mathfrak{n}_i = \{x \in T : v_i(x) \geq 1\}, \quad \mathfrak{m}^r = \pi^r T = \{x \in T : v_0(x) \geq r, v_1(x) \geq r\}.$$

(The last identity uses $\mathfrak{m} = \mathfrak{n}_0 \cap \mathfrak{n}_1$ and that T is a PID, so $\mathfrak{m}^r = \pi^r T$.)

2.2 A difference lemma

For $N \geq 1$ set

$$u_N = t(t+1)^{N+1} = \pi(t+1)^N \in \mathfrak{m} \subseteq D, \quad w_N = (t+1)t^{N+1} = \pi t^N \in \mathfrak{m} \subseteq D.$$

Their valuations are

$$v_0(u_N) = 1, \quad v_1(u_N) = N+1, \quad v_0(w_N) = N+1, \quad v_1(w_N) = 1.$$

Lemma 2.3. *Let $H \subseteq R$ be a finite set of integer-valued polynomials. Then for all sufficiently large N ,*

$$h(u_N) - h(0) \in \mathfrak{m} \quad \text{and} \quad h(w_N) - h(0) \in \mathfrak{m} \quad (h \in H). \quad (2.1)$$

Proof. We prove the statement for u_N ; the proof for w_N is identical with the roles of v_0, v_1 (equivalently $t, t+1$) interchanged. Write each $h \in H$ as $h(X) = \sum_{r \geq 0} c_r X^r$ with $c_r \in K = \mathbb{F}_2(t)$. The finite set H has only finitely many nonconstant coefficients c_r ($r \geq 1$), so we may choose N large enough that

$$v_1(c_r) \geq -(N-1)$$

for every nonconstant coefficient c_r of every $h \in H$. Since $v_1(u_N) = N+1$, for each $r \geq 1$

$$v_1(c_r u_N^r) \geq -(N-1) + r(N+1) \geq -(N-1) + (N+1) = 2.$$

Therefore

$$h(u_N) - h(0) = \sum_{r \geq 1} c_r u_N^r$$

has v_1 -value at least 2, so in particular it lies in $\mathfrak{n}_1$. On the other hand $u_N, 0 \in D$ and $h \in R = \text{Int}(D)$, so $h(u_N), h(0) \in D$ and hence $h(u_N) - h(0) \in D$. Combining, $h(u_N) - h(0) \in \mathfrak{n}_1 \cap D = \mathfrak{m}$ by Proposition 2.2(1). This is (2.1). $\square$

2.3 Failure of injectivity

Let

$$p(X) = X^2 + X.$$

Over $D/\mathfrak{m} \cong \mathbb{F}_2$ we have $0^2 + 0 = 0$ and $1^2 + 1 = 0$, so $p(D) \subseteq \mathfrak{m}$. Since $\mathfrak{m}$ is an ideal of T and the leading manipulations stay inside $K[X]$, multiplying p by any $c \in T$ keeps its values in $\mathfrak{m}$:

$$cp \in R \quad \text{for every } c \in T, \quad (2.2)$$

because $(cp)(D) \subseteq c\mathfrak{m} \subseteq \mathfrak{m} \subseteq D$. In particular $p, tp, (t+1)p \in R$.

Lemma 2.4. *The three polynomials $p, tp, (t+1)p$ all lie outside $\mathfrak{m}R$. Consequently their classes $\bar{p}$ and $\overline{tp}$ are linearly independent over $D/\mathfrak{m} \cong \mathbb{F}_2$ in the $\mathbb{F}_2$ -vector space $R/\mathfrak{m}R$.*

Proof. Suppose, for contradiction, that $cp \in \mathfrak{m}R$ for some $c \in \{1, t, t+1\}$, say

$$cp = \sum_{i=1}^r a_i h_i, \quad a_i \in \mathfrak{m}, h_i \in R. \quad (2.3)$$

First take $c \in \{1, t+1\}$. By Lemma 2.3 applied to $H = \{h_1, \dots, h_r\}$, choose N with $h_i(u_N) - h_i(0) \in \mathfrak{m}$ for all i , and put $u = u_N$. Evaluating (2.3) at u and at 0 (note $p(0) = 0$) and subtracting,

$$c(u^2 + u) = cp(u) = \sum_{i=1}^r a_i(h_i(u) - h_i(0)) \in \mathfrak{m} \cdot \mathfrak{m} = \mathfrak{m}^2. \quad (2.4)$$

We compute the v_0 -value of the left side. We have $v_0(u) = 1$ and $v_0(c) = 0$ (as $c \in \{1, t+1\}$ is a unit at $\mathfrak{n}_0$), while $1+u$ is a unit of T because $u \in \mathfrak{m} \subseteq \mathfrak{n}_0$, so $v_0(1+u) = 0$. Hence

$$v_0(c(u^2 + u)) = v_0(cu(1+u)) = 0 + 1 + 0 = 1.$$

But every nonzero element of $\mathfrak{m}^2 = \pi^2 T$ has v_0 -value ≥ 2 . This contradicts (2.4).

Now take $c = t$. Repeat the argument with $w = w_N$ in place of u and v_1 in place of v_0 . Here $v_1(w) = 1$, $v_1(t) = 0$, and $v_1(1+w) = 0$ (since $w \in \mathfrak{m} \subseteq \mathfrak{n}_1$), so $v_1(t(w^2 + w)) = 1$, again contradicting membership in $\mathfrak{m}^2$. This proves $p, tp, (t+1)p \notin \mathfrak{m}R$.

For the independence statement: a nontrivial $\mathbb{F}_2$ -linear combination of $\bar{p}$ and $\bar{tp}$ equals the class of exactly one of p, tp , or $p+tp = (t+1)p$, none of which lies in $\mathfrak{m}R$. Hence no such combination is zero in $R/\mathfrak{m}R$, i.e. $\bar{p}, \bar{tp}$ are independent. $\square$

Theorem 2.5. *The map $\theta_2 : \text{Int}(D)^{\otimes D^2} \rightarrow \text{Int}(D^2)$ is not injective. Explicitly,*

$$\tau = (tp) \otimes p - p \otimes (tp) \in R \otimes_D R$$

is a nonzero element of $\ker \theta_2$; moreover $\mathfrak{m}\tau = 0$, so $R \otimes_D R$ has nonzero $\mathfrak{m}$ -torsion.

Proof. $\tau \neq 0$. Reducing modulo $\mathfrak{m}$, the natural map $R \otimes_D R \rightarrow (R/\mathfrak{m}R) \otimes_{D/\mathfrak{m}} (R/\mathfrak{m}R) = (R/\mathfrak{m}R) \otimes_{\mathbb{F}_2} (R/\mathfrak{m}R)$ sends τ to

$$\bar{tp} \otimes \bar{p} - \bar{p} \otimes \bar{tp}.$$

By Lemma 2.4, $\bar{p}$ and $\bar{tp}$ are $\mathbb{F}_2$ -linearly independent, so $\bar{p}, \bar{tp}$ extend to a basis of $R/\mathfrak{m}R$ and the two pure tensors $\bar{tp} \otimes \bar{p}$ and $\bar{p} \otimes \bar{tp}$ are distinct basis vectors of the tensor product. Their difference is therefore nonzero, whence $\tau \neq 0$.

$\theta_2(\tau) = 0$. Directly,

$$\theta_2(\tau) = (tp)(X)p(Y) - p(X)(tp)(Y) = tp(X)p(Y) - tp(X)p(Y) = 0,$$

since the scalar $t \in K$ commutes past the polynomials. (Both summands are genuinely elements of $\text{Int}(D^2)$: θ_2 is a D -algebra map and the individual products $f_i(X)h_i(Y)$ lie in $\text{Int}(D^2)$.)

$\mathfrak{m}$ -torsion. We claim $\pi\tau = 0$. Both π and πt lie in D (indeed $\pi = t(t+1) \in \mathfrak{m}$ and $\pi t = t^2(t+1) \in \mathfrak{m}$, while $\mathfrak{m} \subseteq D$), so they may be transferred across $\otimes_D$. Using $\pi(tp) = (\pi t)p$ with $\pi t \in D$, and $\pi \in D$,

$$\pi\tau = (\pi t p) \otimes p - (\pi p) \otimes (tp) = p \otimes (\pi t p) - p \otimes (\pi t p) = 0,$$

since $(\pi t p) \otimes p = p \otimes (\pi t p)$ (moving $\pi t \in D$ across) and $(\pi p) \otimes (tp) = p \otimes (\pi tp) = p \otimes (\pi t p)$ (moving $\pi \in D$ across). For any $a \in \mathfrak{m}$, since $\mathfrak{m}$ is a T -ideal, both $a \in D$ and $at \in \mathfrak{m} \subseteq D$. Then

$$a\tau = (at \cdot p) \otimes p - (a \cdot p) \otimes (tp) = p \otimes (at \cdot p) - p \otimes (at \cdot p) = 0.$$

$\square$

2.4 Failure of surjectivity

Define

$$q(X) = \frac{X^2 + X}{\pi} \in K[X], \quad g(X) = q(X)^2 + q(X) \in K[X].$$

Lemma 2.6. *$g \in R = \text{Int}(D)$ and $g(D) \subseteq \mathfrak{m}$ with $g(0) = 0$. Consequently the two-variable polynomial $P(X, Y) = g(XY)$ lies in $\text{Int}(D^2)$.*

Proof. Let $x \in D$. Then $x^2 + x = p(x) \in \mathfrak{m} = \pi T$ (as in §2.3), so $q(x) = (x^2 + x)/\pi \in T$. Both residue fields $T/\mathfrak{n}_0, T/\mathfrak{n}_1$ are $\mathbb{F}_2$, so for any $y \in T$ the element $y^2 + y$ vanishes modulo each $\mathfrak{n}_i$; hence $y^2 + y \in \mathfrak{n}_0 \cap \mathfrak{n}_1 = \mathfrak{m}$. Applying this to $y = q(x) \in T$ gives

$$g(x) = q(x)^2 + q(x) \in \mathfrak{m} \quad (x \in D),$$

so $g(D) \subseteq \mathfrak{m} \subseteq D$ and $g \in R$. Clearly $g(0) = q(0)^2 + q(0) = 0$. Finally, for $(x, y) \in D^2$ we have $xy \in D$, so $P(x, y) = g(xy) \in \mathfrak{m} \subseteq D$; thus $P \in \text{Int}(D^2)$. $\square$

Theorem 2.7. *The polynomial $P(X, Y) = g(XY)$ lies in $\text{Int}(D^2)$ but not in $\text{im } \theta_2$. Hence $\theta_2 : \text{Int}(D)^{\otimes D^2} \rightarrow \text{Int}(D^2)$ is not surjective.*

Proof. By Lemma 2.6, $P \in \text{Int}(D^2)$. Suppose, for contradiction, that $P \in \text{im } \theta_2$, say

$$P(X, Y) = g(XY) = \sum_{i=1}^r f_i(X) h_i(Y), \quad f_i, h_i \in R. \quad (2.5)$$

Apply Lemma 2.3 to the finite set $H = \{f_1, \dots, f_r, h_1, \dots, h_r\}$ and choose N accordingly; put $u = u_N = \pi(t+1)^N$. Form the second mixed difference of (2.5),

$$\Delta := P(u, u) - P(u, 0) - P(0, u) + P(0, 0).$$

On the right of (2.5) the mixed difference factors:

$$\Delta = \sum_{i=1}^r (f_i(u) - f_i(0)) (h_i(u) - h_i(0)) \in \mathfrak{m} \cdot \mathfrak{m} = \mathfrak{m}^2, \quad (2.6)$$

each factor lying in $\mathfrak{m}$ by Lemma 2.3. On the left, $P(X, Y) = g(XY)$ and $g(0) = 0$ give $P(u, 0) = P(0, u) = P(0, 0) = g(0) = 0$, while $P(u, u) = g(u^2)$. Hence

$$\Delta = g(u^2) \in \mathfrak{m}^2. \quad (2.7)$$

We now contradict (2.7) by a direct valuation computation. Write $z = (t+1)^N$, so $u = \pi z$ and $u^2 = \pi^2 z^2$, $u^4 = \pi^4 z^4$. Then

$$q(u^2) = \frac{u^4 + u^2}{\pi} = \frac{\pi^4 z^4 + \pi^2 z^2}{\pi} = \pi^3 z^4 + \pi z^2.$$

Since $v_0(z) = v_0(t+1) = 0$, the two summands have v_0 -values $v_0(\pi^3 z^4) = 3$ and $v_0(\pi z^2) = 1$; the minimum is attained uniquely, so

$$v_0(q(u^2)) = 1, \quad \text{i.e.} \quad q(u^2) \in \mathfrak{m} \setminus \mathfrak{m}^2.$$

Therefore $v_0(q(u^2)^2) = 2$, while $v_0(q(u^2)) = 1$, so

$$v_0(g(u^2)) = v_0(q(u^2)^2 + q(u^2)) = \min\{2, 1\} = 1.$$

Every nonzero element of $\mathfrak{m}^2 = \pi^2 T$ has v_0 -value ≥ 2 ; as $v_0(g(u^2)) = 1$ we get $g(u^2) \notin \mathfrak{m}^2$, contradicting (2.7). Hence no decomposition (2.5) exists, i.e. $P \notin \text{im } \theta_2$, and θ_2 is not surjective. $\square$

2.5 All $n \geq 2$

We deduce Theorem 2.1 from the $n = 2$ failures by padding (for injectivity) and restriction (for surjectivity).

Proposition 2.8. *For every $n \geq 2$, the element*

$$\tau_n := \tau \otimes \underbrace{1 \otimes \cdots \otimes 1}_{n-2} \in R^{\otimes D^n}$$

is a nonzero element of $\ker \theta_n$, where $\tau = (tp) \otimes p - p \otimes (tp)$ is as in Theorem 2.5. Hence θ_n is not injective.

Proof. $\tau_n \in \ker \theta_n$. Applying θ_n ,

$$\theta_n(\tau_n) = \left(tp(X_1)p(X_2) - tp(X_1)p(X_2) \right) \cdot 1 \cdots 1 = 0,$$

exactly as in Theorem 2.5, the variables $X_3, \dots, X_n$ contributing the constant polynomial 1.

$\tau_n \neq 0$. Evaluation at 0 gives a D -algebra homomorphism $\text{ev}_0 : R \rightarrow D$, $f \mapsto f(0)$ (well defined since $0 \in D$), with $\text{ev}_0(1) = 1$. The composite

$$R \otimes_D R \xrightarrow{\xi \mapsto \xi \otimes 1^{\otimes (n-2)}} R^{\otimes D^n} \xrightarrow{\text{id} \otimes \text{ev}_0^{\otimes (n-2)}} R \otimes_D R$$

is the identity, because $\text{ev}_0(1) = 1$. Thus $\xi \mapsto \xi \otimes 1^{\otimes (n-2)}$ is a split injection, and $\tau_n = \tau \otimes 1^{\otimes (n-2)} \neq 0$ since $\tau \neq 0$ by Theorem 2.5. $\square$

Proposition 2.9. *For every $n \geq 2$, the polynomial $P(X_1, X_2) = g(X_1 X_2) \in \text{Int}(D^n)$ (viewed as independent of $X_3, \dots, X_n$) does not lie in $\text{im } \theta_n$. Hence θ_n is not surjective.*

Proof. $P \in \text{Int}(D^n)$ since it already lies in $\text{Int}(D^2)$ by Lemma 2.6 and does not involve $X_3, \dots, X_n$. Suppose $P \in \text{im } \theta_n$, say

$$g(X_1 X_2) = \sum_i f_{i,1}(X_1) f_{i,2}(X_2) \cdots f_{i,n}(X_n), \quad f_{i,j} \in R.$$

Specialize $X_3 = \cdots = X_n = 0$. Each $f_{i,j}(0) \in D$ for $j \geq 3$, so writing $\lambda_i = \prod_{j \geq 3} f_{i,j}(0) \in D$ we obtain

$$g(X_1 X_2) = \sum_i (\lambda_i f_{i,1}(X_1)) f_{i,2}(X_2),$$

with $\lambda_i f_{i,1} \in R$ (as R is a D -algebra). This exhibits $g(X_1 X_2) \in \text{im } \theta_2$, contradicting Theorem 2.7. Hence $P \notin \text{im } \theta_n$. $\square$

Proof of Theorem 2.1. The domain $D = \mathbb{F}_2 + \mathfrak{m}$ of §2.1 is a one-dimensional Noetherian local integral domain by Proposition 2.2. For every $n \geq 2$, Proposition 2.8 shows θ_n is not injective and Proposition 2.9 shows θ_n is not surjective. In particular θ_n is never an isomorphism. (For $n = 2$ these are Theorems 2.5 and 2.7.) $\square$

References

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